

ZAIN BUGHIO

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EDUCATION

University of Waterloo, B.Math - Double Major in Applied Math (Scientific ML) & Statistics

Expected 2029

SKILLS

Programming Languages

Java, Python, JavaScript, CSS, SQL, C, TypeScript, HTML, Racket

Machine Learning

MLFlow, Scikit-learn, ONNX, Pytorch,

Frameworks and Libraries

React, Material UI, FastAPI, Pandas, Numpy, Tailwind CSS, Streamlit

Tools and Platforms

GitHub, Jira, PowerShell, Linux, VS Code, Vercel, Tableau, Supabase

Soft Skills

Collaboration, Time Management, Problem Solving, Team Work, Organization

EXPERIENCE

Software Engineer

May 2025 - December 2025

PixelsBoost

Milton, ON

- Engineered and launched **6 production-grade full-stack websites** using **React, HTML/CSS, and JavaScript**, featuring dynamic contact forms, interactive image galleries, and fully responsive navigation, collectively serving **1,500+ monthly active users**.
- Architected seamless integrations with **3 mission-critical third-party APIs (Stripe, Google Maps, SendGrid)**, enabling **\$10K+ USD** in monthly payment processing and automated email workflows while sustaining **99%** uptime through robust error handling.
- Spearheaded delivery of **5 simultaneous client projects** in an **agile environment**, authoring **200+ commits** across **20+ feature branches**, mentoring cross-functional collaboration with 2 designers, and maintaining code integrity through resolution of **15+ merge conflicts**.
- Drove a **40%** performance uplift across client websites by implementing WebP compression, lazy loading, and **Cloudflare CDN**, elevating Google Lighthouse scores from 65 → 86 and cutting bounce rates by **15%**

Information Technology Intern

July 2024 - August 2024

CDI College

Mississauga, ON

- Installed, configured, and maintained desktops, laptops, printers, and mobile devices using **Intune**.
- Troubleshoot hardware, software, and peripheral issues, ensuring minimal downtime for over **500 students/staff**.
- Provided responsive technical assistance, contributing to seamless daily IT operations.

PROJECTS

Wanderers(Hack Canada) | Next.js, Supabase, Gemini, Cloudinary, Fast-API, Scikit-learn, Cloudinary, Vercel

- Architected and shipped a full-stack campus social app in **Next.js 15** using **Supabase Auth, Google Maps, and Tailwind CSS**, featuring real-time activity bubbles with join flows and an End Event flow with live camera capture.
- Built a Moments feed with likes, comments, and **Web Share API**, enabling seamless content sharing and social interaction across campus activity events.
- Integrated **Google Gemini** for server-side intent parsing, **Cloudinary** for media uploads, and a **K-Means clustering recommendation model (K=6, 72.2% Hit Rate@6)** trained with **scikit-learn** using tuned score weights across cluster popularity, interest match, time match, and proximity.
- Designed **REST API** routes and backend **schemas** covering auth, bubbles, moments, connections, and chat unlock logic.

CoopTrack | React.JS, Typescript, PostgreSQL, Supabase, Tailwind CSS, Vercel, Gemini

- Engineered a full-stack SPA using **React 18, TypeScript, and Supabase PostgreSQL** with a normalized schema supporting **250,000+ entries** and dynamic query filtering by user and co-op term.
- Implemented JWT-based **Google OAuth 2.0**, row-level security for multi-tenant data isolation, and automated **CI/CD deployment** on **Vercel**, acquiring **30+ users** within 24 hours of launch.
- Built a mobile-responsive component library with **Tailwind CSS** and **shadcn/ui**, client-side routing via React Router, and optimistic UI updates supporting **1,000+ concurrent users**.
- Integrated the **Gemini AI API** to enable intelligent form autofill, allowing users to paste a job description and automatically populate all relevant application fields through structured **LLM output parsing**, reducing manual data entry.